

Python Lab Viva Quiz

Programming Languages Lab (Python) - F.Y. Sem I
Experiments 1 to 5 - 50 Multiple Choice Questions

Name: _____

Roll No: _____

Batch / Div: _____

Date: _____

Instructions: Choose the one correct option for each question. Each quiz has 10 questions.

Quiz 1: Sum, Average, Minimum and Maximum

Q1. What data type does input() return?

- a) int
- b) float
- c) str
- d) bool

Q2. What is the sum of 12, 45, 7, 89 and 23?

- a) 166
- b) 176
- c) 186
- d) 196

Q3. Which formula gives the average?

- a) $\text{Sum} \times n$
- b) Sum / n
- c) n / Sum
- d) $\text{Sum} - n$

Q4. Why are minimum and maximum set to None at the start?

- a) Because no number has been entered yet
- b) Because None is the smallest number
- c) Because Python needs it for loops
- d) To save memory

Q5. Which loop suits a case where we know exactly how many numbers will be entered?

- a) while True
- b) for
- c) do-while
- d) No loop is needed

Q6. Which statement updates the maximum?

- a) if num < maximum:
- b) if num > maximum:
- c) if num == maximum:
- d) if num != maximum:

Q7. For the numbers 4, 6, 10 and 20, what is the average?

- a) 8
- b) 9
- c) 10
- d) 12

Q8. What error occurs if total / count is calculated when count = 0?

- a) TypeError
- b) ValueError
- c) ZeroDivisionError
- d) NameError

Q9. What does float("5") return?

- a) "5"
- b) 5
- c) 5.0
- d) Error

Q10. What is the minimum of 12, 45, 7, 89 and 23?

- a) 12
- b) 7
- c) 23
- d) 89

Quiz 2: Commission, Simple Interest and Compound Interest

Q1. What is the commission formula?

- a) $(\text{Rate} + \text{Sales}) / 100$
- b) $(\text{Rate} \times \text{Sales}) / 100$
- c) $\text{Rate} / \text{Sales}$
- d) $\text{Sales} / \text{Rate}$

Q2. What is the commission on Rs. 20,000 sales at a 3% rate?

- a) Rs. 300
- b) Rs. 600
- c) Rs. 900
- d) Rs. 6,000

Q3. What is the simple interest for $P = 2000$, $R = 5\%$, $T = 3$ years?

- a) Rs. 100
- b) Rs. 200
- c) Rs. 300
- d) Rs. 400

Q4. Which operator is used for exponentiation in Python?

- a) ^
- b) **
- c) //
- d) %

Q5. In $A = P \times (1 + R/100)^T$, what does A represent?

- a) Interest only
- b) Rate
- c) Final amount (principal + interest)
- d) Time

Q6. What is the compound interest for $P = 1000$, $R = 10\%$, $T = 2$ years?

- a) Rs. 200
- b) Rs. 210
- c) Rs. 220
- d) Rs. 121

Q7. When $T = 1$ year, which statement is true?

- a) SI is greater than CI
- b) CI is greater than SI
- c) SI equals CI
- d) Both are zero

Q8. For $T > 1$, how do SI and CI compare?

- a) SI is greater
- b) CI is greater
- c) They are always equal
- d) It depends on the input type

Q9. Why must user inputs be converted using float()?

- a) Because input() returns a string
- b) Because Python has no integers
- c) To make the program faster
- d) To avoid using loops

Q10. What does 5^2 give in Python?

- a) 25
- b) 10
- c) 7
- d) 3

Quiz 3: Student Marks, Grades and Result

Q1. What is the percentage if the total is 395 out of 500?

- a) 75%
- b) 79%
- c) 80%
- d) 85%

Q2. Which grade does a student with 79.5% get?

- a) A
- b) B
- c) C
- d) D

Q3. Which grade does exactly 90% get?

- a) A
- b) A+
- c) B
- d) E

Q4. Why are grade conditions checked from highest to lowest?

- a) It looks neater
- b) elif stops at the first true condition, so the wrong order gives wrong grades
- c) Python requires it
- d) It saves memory

Q5. What is the purpose of fail_flag?

- a) To count the subjects
- b) To track whether the student failed in any individual subject
- c) To store the grade
- d) To store the percentage

Q6. A student has 45% overall but scores 30 in one subject (pass mark 40). What is the result?

- a) Pass
- b) Fail
- c) Grade A
- d) Not defined

Q7. A student takes 4 subjects (each out of 100) and gets a total of 300. What is the percentage?

- a) 60%
- b) 70%
- c) 75%
- d) 80%

Q8. Which grade is given for a percentage below 40?

- a) E
- b) D
- c) F
- d) C

Q9. Which keyword is used to check additional conditions after if?

- a) else if
- b) elseif
- c) elif
- d) then

Q10. What is the data type of fail_flag?

- a) int
- b) str
- c) float
- d) bool

Quiz 4: Prime, Armstrong, Factorial, Square and Cube

Q1. Which of these is a prime number?

- a) 1
- b) 9
- c) 11
- d) 15

Q2. Which is the only even prime number?

- a) 0
- b) 2
- c) 4
- d) 6

Q3. Up to which value do we check divisors when testing for a prime?

- a) n
- b) $n / 2$
- c) square root of n
- d) $n - 1$

Q4. What is the value of $10 \% 3$?

- a) 3
- b) 1
- c) 0
- d) 3.33

Q5. What is the value of $17 // 5$?

- a) 3
- b) 3.4
- c) 2
- d) 4

Q6. Which of these is NOT an Armstrong number?

- a) 153
- b) 370
- c) 371
- d) 123

Q7. What is $5!$ (5 factorial)?

- a) 25
- b) 100
- c) 120
- d) 720

Q8. What is $0!$?

- a) 0
- b) 1
- c) Undefined
- d) Infinity

Q9. Why is factorial initialized as 1 and not 0?

- a) Multiplying by 0 would always give 0
- b) 1 is faster
- c) Python requires it
- d) It makes the loop shorter

Q10. What is the cube of 4?

- a) 12
- b) 16
- c) 64
- d) 256

Quiz 5: String Operations

Q1. What is len("Python 3")?

- a) 6
- b) 7
- c) 8
- d) 9

Q2. What is "Python".find('o')?

- a) 3
- b) 4
- c) 5
- d) -1

Q3. What does find() return when the substring is not found?

- a) 0
- b) None
- c) -1
- d) Error

Q4. What is "Hello"[::-1]?

- a) Hello
- b) olleH
- c) HELLO
- d) hello

Q5. What is "Mississippi".count('s')?

- a) 2
- b) 3
- c) 4
- d) 5

Q6. What is "banana".replace("a", "o")?

- a) bonana
- b) banono
- c) bonono
- d) banana

Q7. What does "Python" == "python" return?

- a) True
- b) False
- c) Error
- d) None

Q8. What is "Hello".upper()?

- a) hello
- b) Hello
- c) HELLO
- d) hELLO

Q9. What does "tho" in "Python" return?

- a) True
- b) False
- c) 2
- d) -1

Q10. What is "Hello" + " " + "World"?

- a) HelloWorld
- b) Hello World
- c) Hello+World
- d) Error